InsureWell

Health Insurance Management System

System Diagrams

Version 1.0 | March 2026
Prepared for: Capstone Project Review

1. Use Case Diagram

The Use Case Diagram below illustrates all actors in the InsureWell system and the use cases available to each role. It covers four primary actors: Customer, Admin, Underwriter, and Claims Officer. The diagram is grouped into five functional domains: Policy Management, Admin Operations, Underwriter Operations, Claims Management, and Authentication.

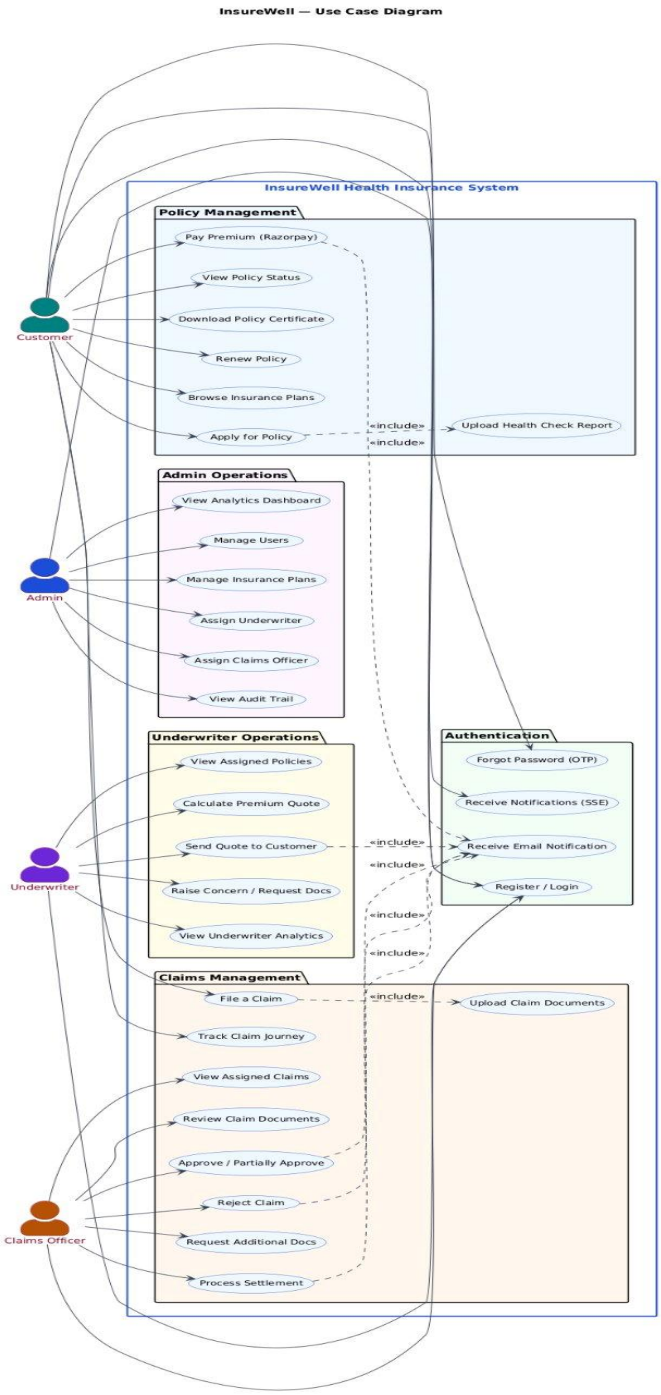

Figure 1 — InsureWell Use Case Diagram

2. Class Diagram

The Class Diagram presents the core domain entities and their relationships in the InsureWell system. It uses single-table inheritance for the User hierarchy and shows how entities such as Policy, Claim, Payment, and InsurancePlan are interconnected. Key enumerations (PolicyStatus, ClaimStatus, PaymentMethod) and auxiliary classes (Notification, AuditLog, RefreshToken) are also shown.

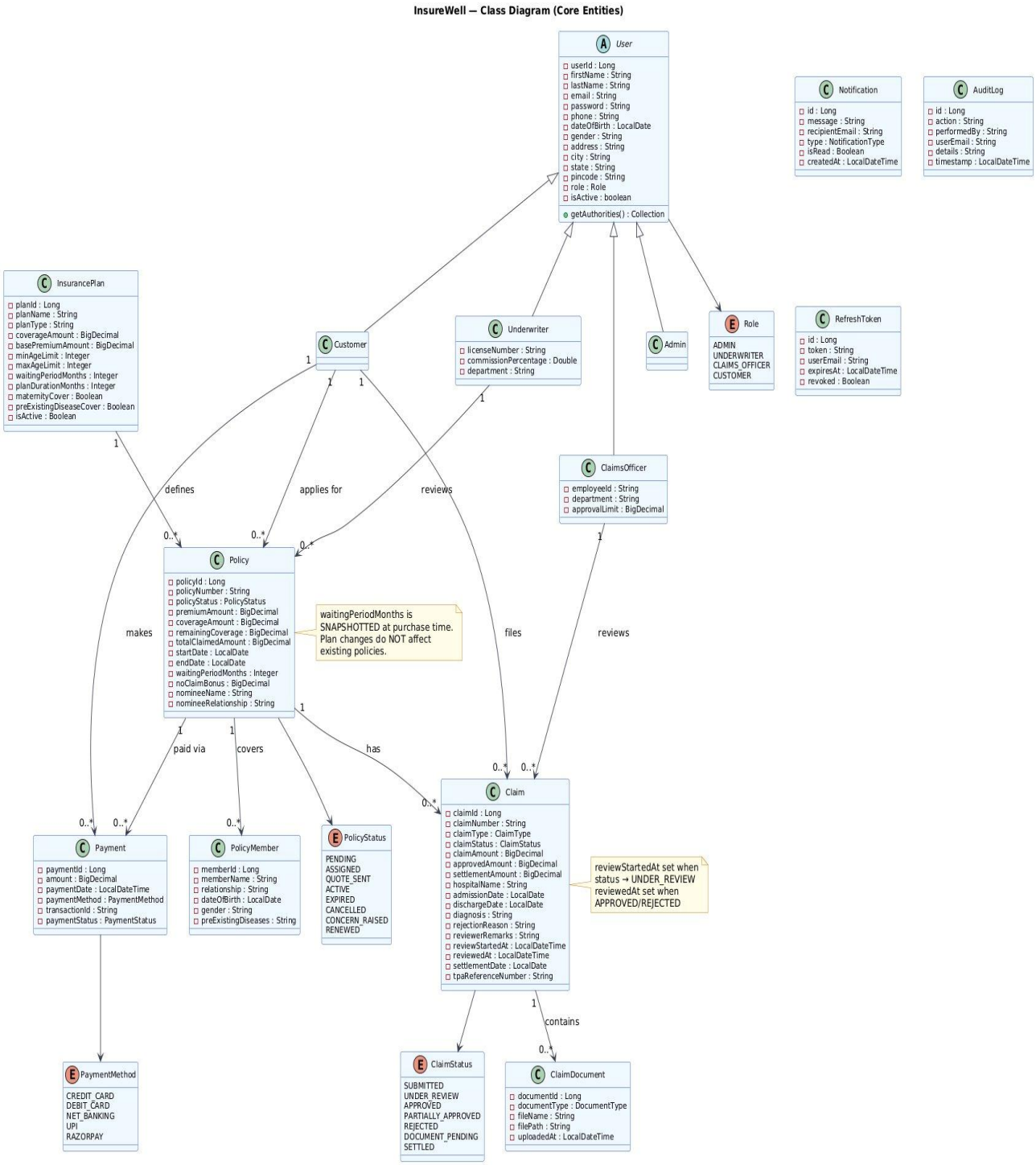

Figure 2 — InsureWell Class Diagram (Core Entities)

3. Entity Relationship (ER) Diagram

The Entity Relationship Diagram defines the complete database schema for InsureWell. It shows all tables, their columns with data types, primary keys, foreign keys, and the cardinality of relationships between entities. The schema covers the USERS, INSURANCE_PLANS, POLICIES, POLICY_MEMBERS, CLAIMS, CLAIM_DOCUMENTS, PAYMENTS, NOTIFICATIONS, AUDIT_LOGS, REFRESH_TOKENS, and PASSWORD_RESET_OTPS tables.

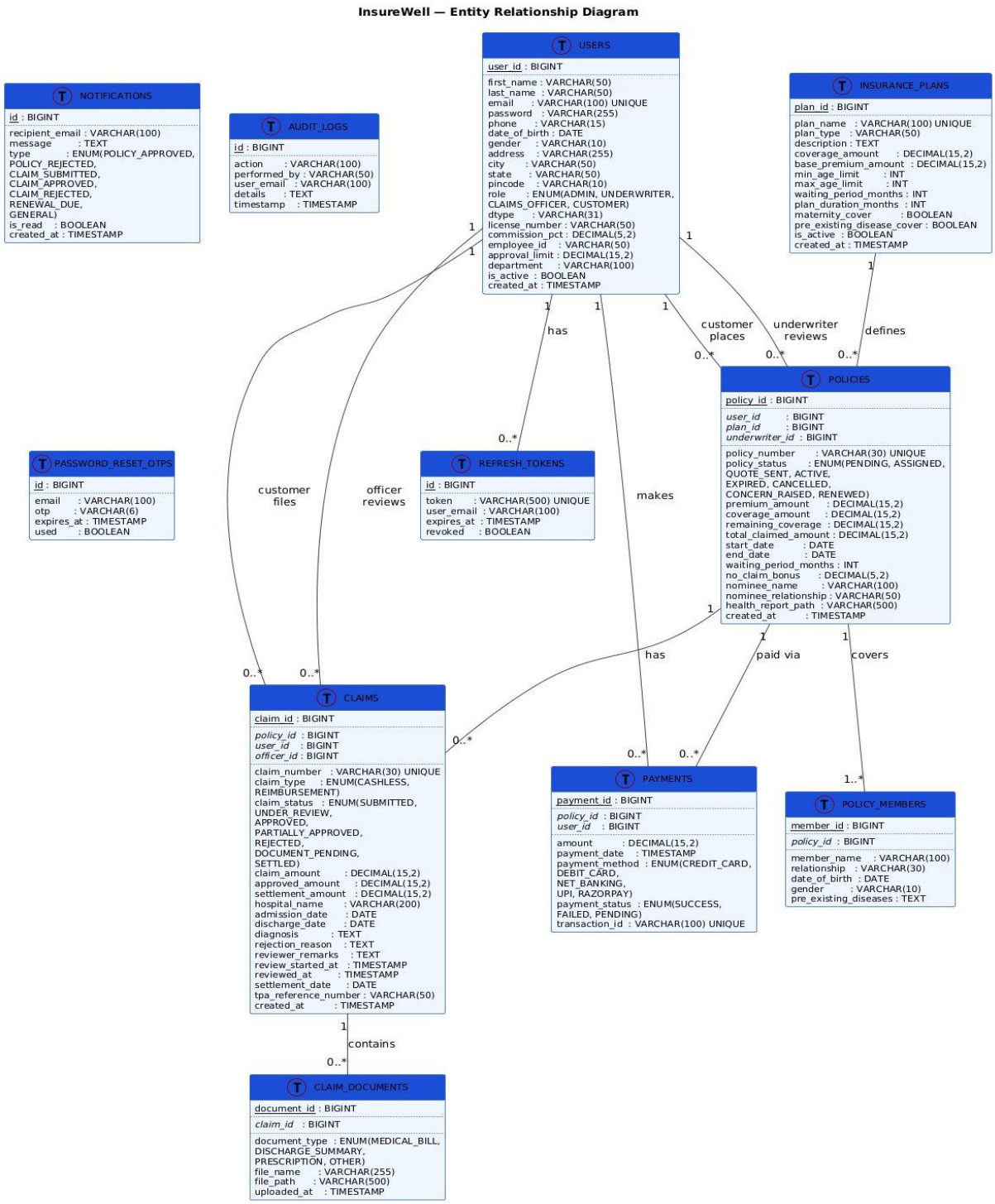

Figure 3 — InsureWell Entity Relationship Diagram

4. Activity Diagrams

Activity diagrams describe the step-by-step workflows for each role in the InsureWell system. Each diagram shows the sequence of actions, decision points, API calls, and status transitions that occur during a user's interaction with the platform.

4.1 Customer Activity Diagram

The Customer Activity Diagram covers the full policy application lifecycle from the customer's perspective: browsing and selecting a plan, filling the application form, adding members with date-of-birth validation, uploading the mandatory health check report, submitting the application, tracking status transitions (PENDING → ASSIGNED → QUOTE_SENT), completing payment via Razorpay, and downloading the policy certificate upon activation.

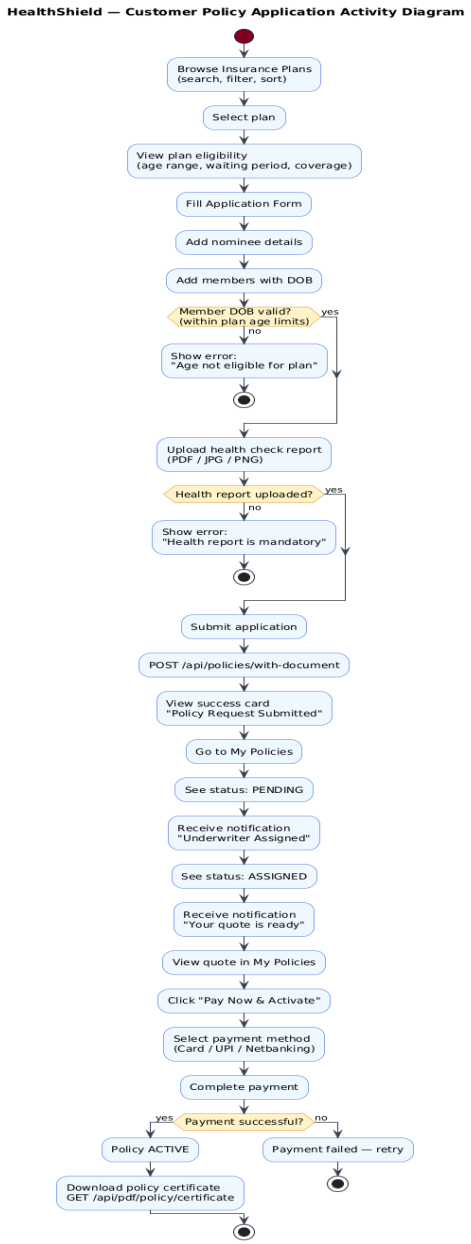

Figure 4 — Customer Policy Application Activity Diagram

4.2 Admin Activity Diagram

The Admin Activity Diagram illustrates the four concurrent administrative workflows: User Management (activate/deactivate accounts), Plan Management (create/update/toggle plans), Policy Assignment (assign underwriters to pending applications), and Claims Assignment (assign claims officers to submitted claims). The diagram uses parallel fork notation to show these workflows operating in parallel.

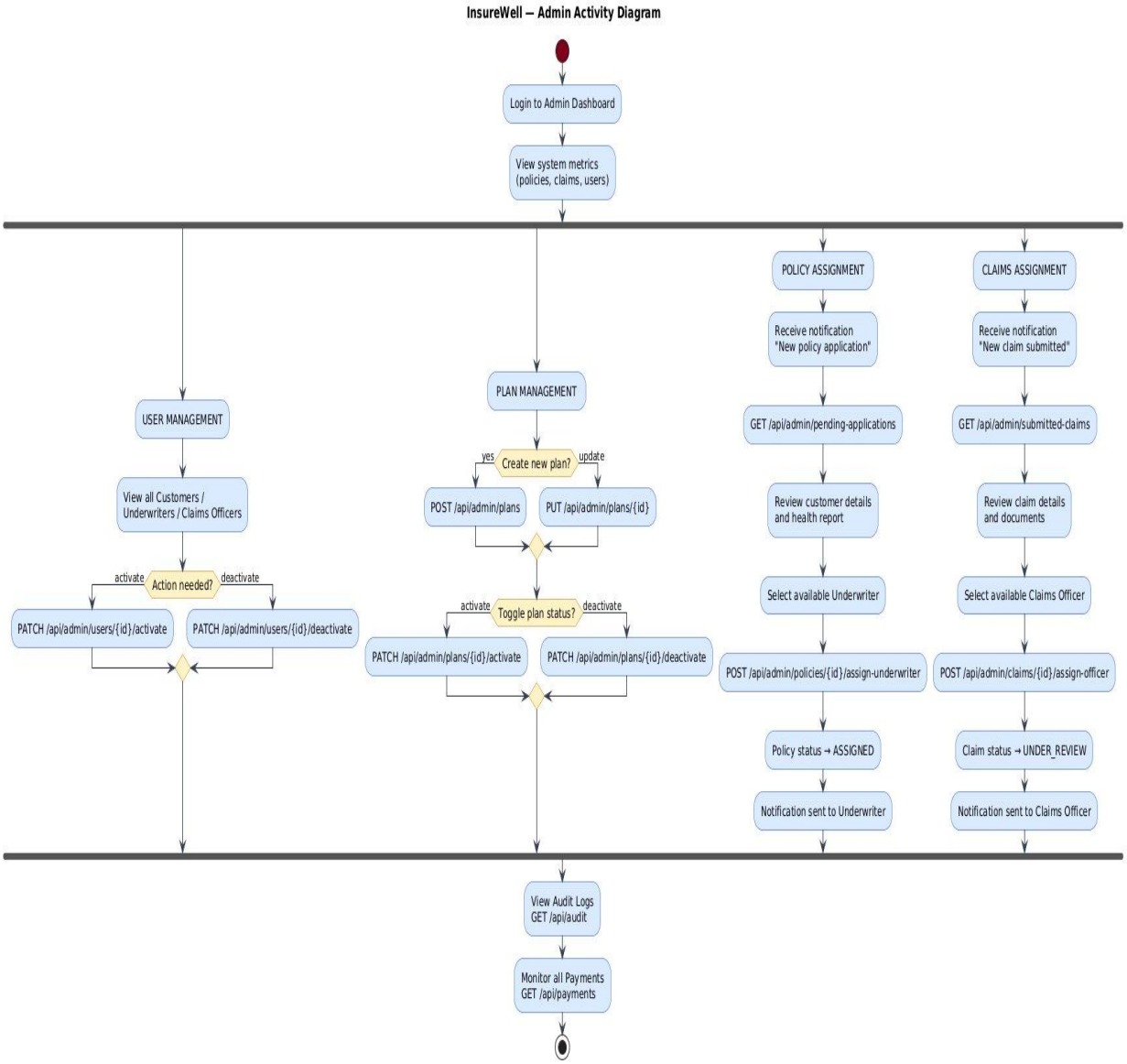

Figure 5 — Admin Activity Diagram

4.3 Underwriter Activity Diagram

The Underwriter Activity Diagram covers the risk assessment workflow: receiving a policy assignment notification, reviewing customer and member details, clicking Calculate Quote (which calls the premium breakdown endpoint), deciding whether to raise a concern or send the final quote. A repeat loop handles multiple assignments sequentially.

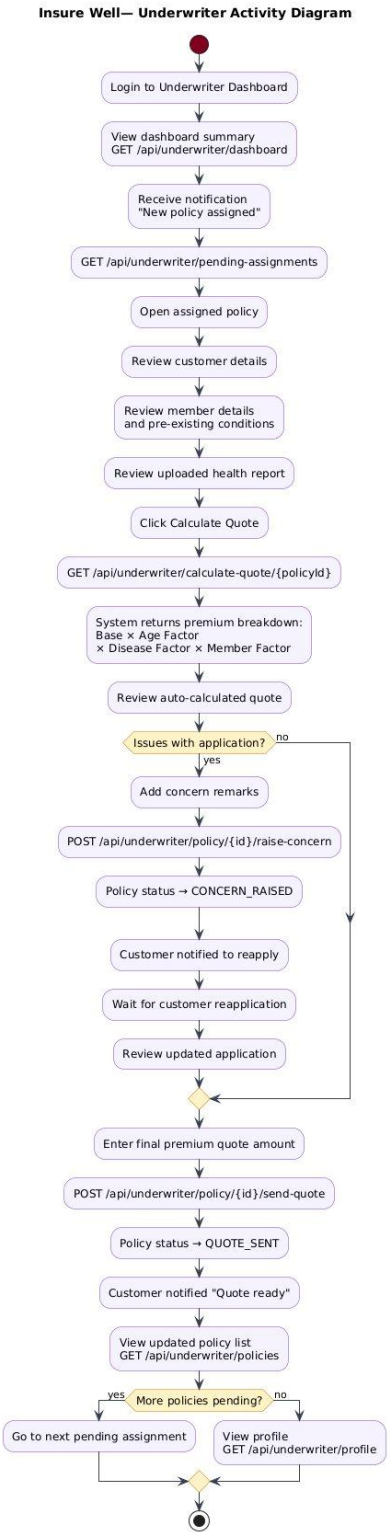

Figure 6 — Underwriter Activity Diagram

4.4 Claims Officer Activity Diagram

The Claims Officer Activity Diagram covers the claim review lifecycle: checking the claims queue, self-assigning (pickup) a claim, reviewing details and verifying documents, handling missing documents (DOCUMENT_PENDING), and making a four-way decision: Approve (triggering full settlement), Partially Approve (triggering partial settlement), Reject (notifying customer with reason), or Escalate (APPEAL_REQUESTED). A repeat loop processes multiple claims in sequence.

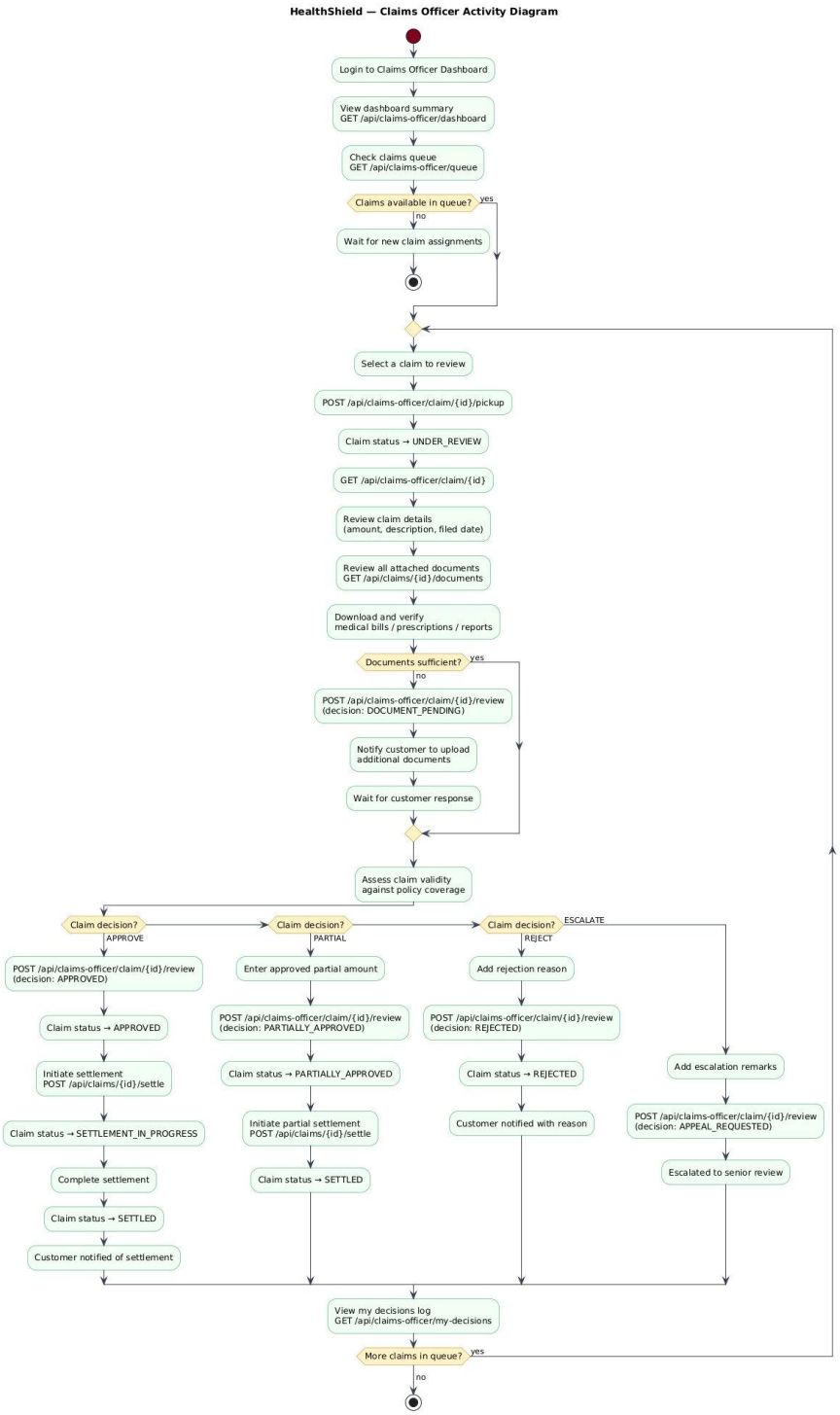

Figure 7 — Claims Officer Activity Diagram

5. Sequence Diagrams

Sequence diagrams show the detailed message flow between system components for the two most critical end-to-end workflows: Policy & Payment Processing and Claims Processing. They illustrate interactions between the Angular Frontend, Spring Boot Controllers, Services, the H2/MySQL Database, Email Service, Notification Service, Razorpay API, and relevant actors.

5.1 Policy and Payment Processing Sequence Diagram

This sequence diagram covers four phases: (1) Authentication — customer login and JWT issuance; (2) Policy Application — submitting the application with health report, saving Policy with status PENDING, and notifying Admin; (3) Admin assigns Underwriter — policy transitions to ASSIGNED, underwriter is notified; (4) Underwriter Sends Quote — premium calculation, quote submission (QUOTE_SENT), customer notification; (5) Razorpay Payment — order creation, checkout modal, HMAC signature verification, policy activation (ACTIVE), email and SSE notification.

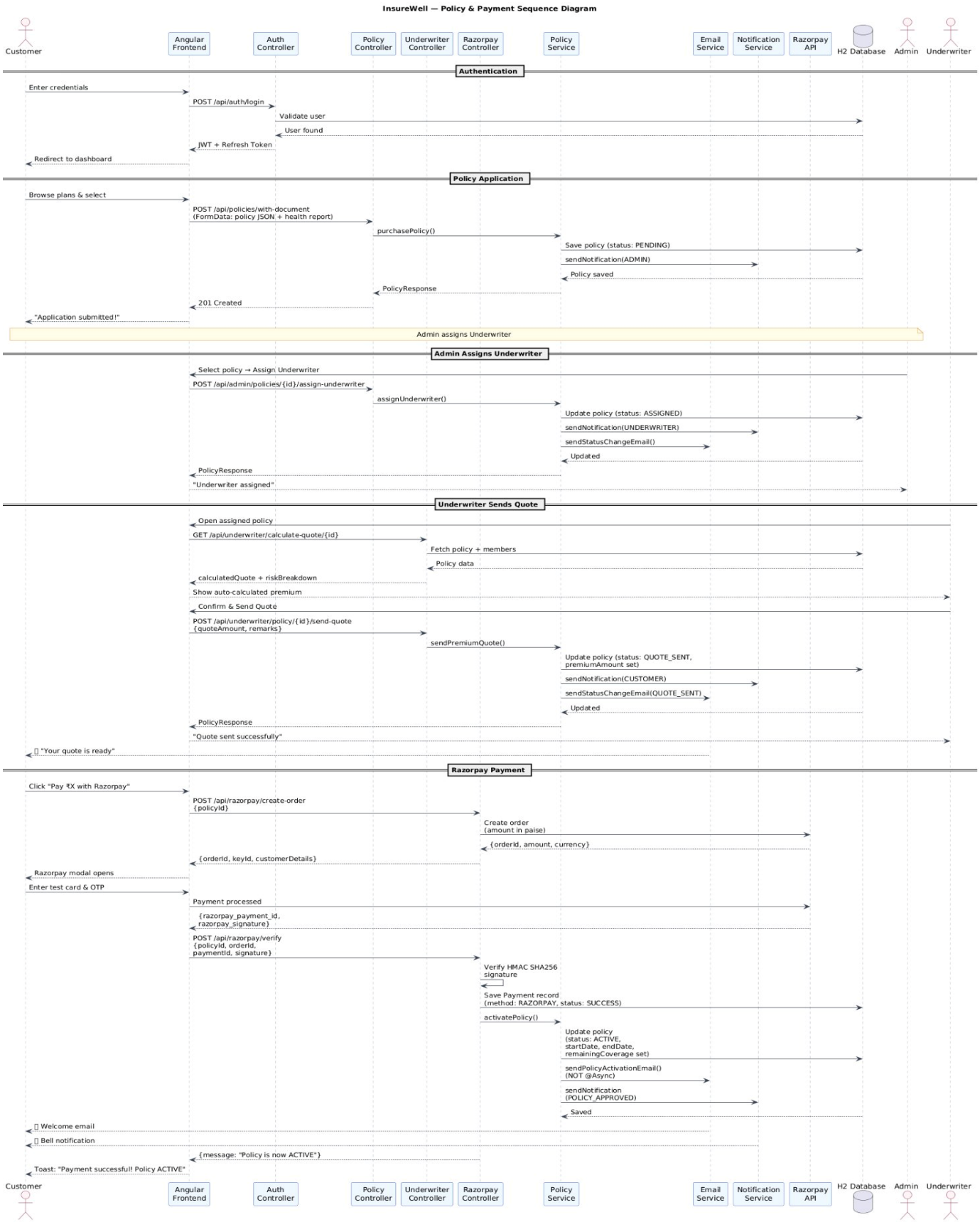

Figure 8 — Policy & Payment Processing Sequence Diagram

5.2 Claims Processing Sequence Diagram

This sequence diagram covers four phases: (1) File a Claim — customer submits multipart claim form with documents; system validates policy ACTIVE status, remaining coverage, and saves Claim (SUBMITTED) and ClaimDocuments; (2) Admin Assigns Claims Officer — officer is notified; (3) Claims Officer Review — officer opens claim, views documents inline, submits review decision (PARTIALLY_APPROVED shown as example), system recalculates remaining coverage and notifies customer; (4) Settlement — officer initiates settlement, system sets SETTLED status, generates TPA reference, updates policy totals, and notifies customer.

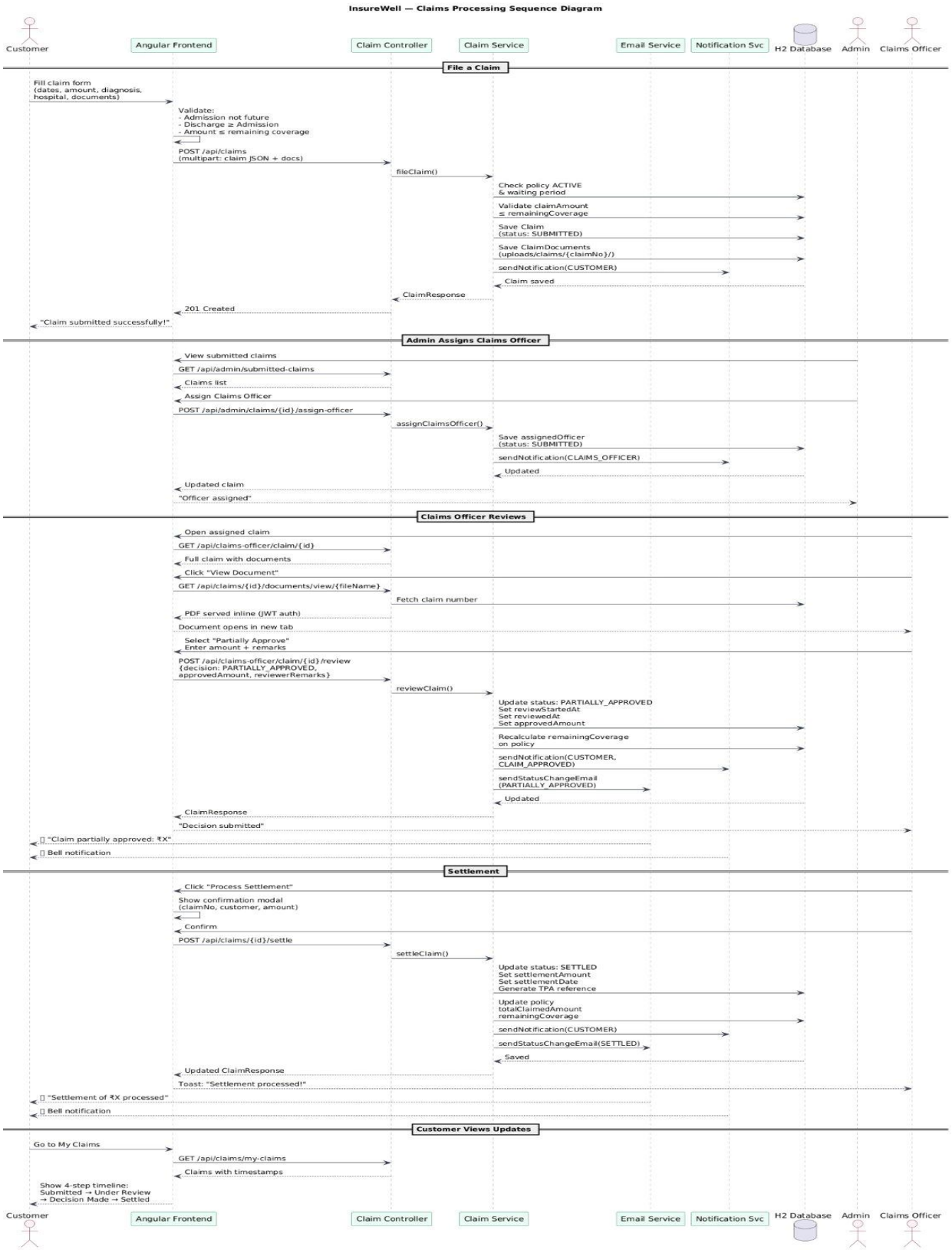

Figure 9 — Claims Processing Sequence Diagram

6. Diagram Index

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5	Activity Diagram — Admin	Section 4.2
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7	Activity Diagram — Claims Officer	Section 4.4
8	Sequence Diagram — Policy & Payment Processing	Section 5.1
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